UW

Department of Computer Science
Assignment 1

Course Code: CS-202
Course Title: Artificial Intelligence
Total Marks: 15
Date: 03-11-2025

Name: ___________________
Registration No. _________________________
Discipline: BS(CS)-3rd B
Submission Date: 06-11-2025

Assignment Instructions

1. **Submission Deadline:**
    - **No late assignments will be accepted after the due date.**
2. **Format Requirements:**
    - **The assignment must be handwritten on A4-size pages.**
    - **Your work should be neat, legible, and well-organized.**
3. **Presentation Guidelines:**
    - **Clearly mention each question (stem) with a proper heading.**
    - **Provide clear justifications or explanations for each answer.**

[CLO1, GA2, C2]

Q1. For the given scenarios **discuss** the following questions:

- Describe the PEAS (Performance measure, Environment, Actuators, and Sensors) components for the scenario.
- Classify the type of environment (e.g., fully observable or partially observable, deterministic or stochastic, etc.).
- Explain the structure and type of the agent (or agents) involved, including how it perceives its environment and decides on actions.

Scenario 1: Smart Home Automation System

Consider a smart home automation system that is designed to manage various aspects of a home, such as lighting, temperature, security, and entertainment. The system uses a variety of sensors and actuators to monitor and control the home environment.

Scenario 2: Autonomous Delivery Drone

Consider an autonomous delivery drone that is used to deliver small packages in an urban environment. The drone is equipped with various sensors and navigation systems to avoid obstacles and navigate to its destination safely.

Scenario 3: AI-Powered Virtual Assistant

Consider an AI-powered virtual assistant designed to help users with tasks such as scheduling meetings, setting reminders, answering questions, and controlling smart devices. The assistant interacts with users through voice or text commands, processes natural language, and provides relevant responses or actions based on user input.

Scenario 4: Industrial Robotic Arm

Consider an industrial robotic arm used in a manufacturing assembly line. The robotic arm performs tasks such as welding, painting, assembling, or packaging products with high precision. It operates using various sensors and actuators to detect objects, manipulate them, and coordinate with other machinery in a factory setting.